Cool math is hiding in "The 12 Days of Christmas"

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1 Introduction

The classic English Christmas carol "The Twelve Days of Christmas" has all kinds of interesting mathematical features. For example, it is an example of a *cumulative song*, in which the lyrics detail a series of increasingly numerous gifts given to the speaker by their "true love" on each of the twelve days of Christmas (the twelve days that make up the Christmas season, starting with Christmas Day) [7]. This cumulative property leads to a bunch of interesting questions including "How many gifts are given each day?" and "What is the total number of gifts given during the 12 Days of Christmas?".

In any event the carol's words were apparently first published in England in the late 18th century (and has a Roud Folk Song Index number of 68 if you happen to be interested in that kind of thing [9]). A large number of different melodies have been associated with the song, of which the best known is derived from a 1909 arrangement of a traditional folk melody by English composer Frederic Austin [10]. There are twelve verses, each describing a gift given by "my true love" on one of the twelve days of Christmas. Suffice it to say that are many variations in the lyrics. That said, the lyrics I'm using here are from Frederic Austin's 1909 publication that established the current form of the carol [8].

The first three verses of Austin's version of the carol are:

1. On the first day of Christmas my true love sent to me: *A partridge in a pear tree.*
2. On the second day of Christmas my true love sent to me: *Two turtle doves, and a partridge in a pear tree.*
3. On the third day of Christmas my true love sent to me: *Three French hens, two turtle doves, and a partridge in a pear tree.*

The subsequent verses follow the same pattern. The verses deal with the next day where it adds one new gift and then repeats all the earlier gifts. So each verse is one line longer than the previous verse. The subsequent verses add

4. 4 calling birds
5. 5 gold rings
6. 6 geese a-laying
7. 7 swans a-swimming
8. 8 maids a-milking
9. 9 ladies dancing
10. 10 lords a-leaping
11. 11 pipers piping
12. 12 drummers drumming

2 How many gifts are received each day and what is the total?

The number of gifts received on day **n** and the total number of gifts received by (including) day **n** are shown in the following table ($1 \leq \mathbf{n} \leq 12$):

Day	Gifts received on day n	Total number of gifts received by day n
1	**1**	**1**
2	1+2=**3**	1+3=**4**
3	1+2+3=**6**	1+3+6=**10**
4	1+2+3+4=**10**	1+3+6+10=**20**
5	1+2+3+4+5=**15**	1+3+6+10+15=**35**
6	1+2+3+4+5+6=**21**	1+3+6+10+15+21=**56**
7	1+2+3+4+5+6+7=**28**	1+3+6+10+15+21+28=**84**
8	1+2+3+4+5+6+7+8=**36**	1+3+6+10+15+21+28+36=**120**
9	1+2+3+4+5+6+7+8+9=**45**	1+3+6+10+15+21+28+36+45=**165**
10	1+2+3+4+5+6+7+8+9+10=**55**	1+3+6+10+15+21+28+36+45+55=**220**
11	1+2+3+4+5+6+7+8+9+10+11=**66**	1+3+6+10+15+21+28+36+45+55+66=**286**
12	1+2+3+4+5+6+7+8+9+10+11+12=**78**	1+3+6+10+15+21+28+36+45+55+66+78=**364**

Table 1: Gift counts for the 12 days of Christmas

Interestingly, the numbers in blue in Table 1, the number of gifts received on day **n**, are called triangular numbers. Why? We will see in Section 4 that there is an interesting geometric interpretation that motivates this name.

The nth triangular number, T_n, is defined as follows:

$$T_n = \sum_{k=1}^{n} k = \frac{n(n+1)}{2} = \binom{n+1}{2} \tag{1}$$

where $\binom{n}{m}$ is the familiar binomial coefficient[1]. For example, $T_{12} = \sum_{k=1}^{12} k = \frac{12 \cdot 13}{2} = \binom{13}{2} = \mathbf{78}$.

So how many total gifts have been received by day **n**? If we look at the **Total number of gifts received by day n** column of Table 1 we can see a pattern. In particular, the number of gifts received by (including) day **n** is the sum of the gifts received on the previous days plus the gifts received on day **n**. The number of gifts received by day **n** is shown in red in Table 1. These numbers are called the tetrahedral numbers and are defined as follows [3]:

$$T_{e_n} = \sum_{k=1}^{n} T_k = \frac{n(n+1)(n+2)}{6} = \binom{n+2}{3} \tag{2}$$

where T_k is the kth triangular number (Equation (1)).

It turns out that the total number of gifts received by day 12, $T_{e_{12}}$, is $\sum_{k=1}^{12} T_k = \frac{12 \cdot 13 \cdot 14}{6} = \binom{14}{3} = \mathbf{364}$.

[1] Aside: The great mathematician Carl Friedrich Gauss is said to have discovered the consecutive integer formula (Equation (1)) while still at primary school [1].

BTW, can a number can be both triangular and tetrahedral? Well, if so there would have to exist $m, n \in \mathbb{N}$ such that

$$T_n = \binom{n+1}{2} = \binom{m+2}{3} = T_{e_m} \tag{3}$$

Apparently the only m, n that satisfy Equation (3) are [4]:

$$\begin{aligned}
T_{e_1} &= T_1 &&= 1 \\
T_{e_3} &= T_4 &&= 10 \\
T_{e_8} &= T_{15} &&= 120 \\
T_{e_{20}} &= T_{55} &&= 1540 \\
T_{e_{34}} &= T_{119} &&= 7140
\end{aligned}$$

3 Pascal's triangle

"The 12 Days of Christmas" also has interesting connections to Pascal's triangle [5]. Specifically, the 3rd diagonal of Pascal's triangle, shown in blue in Figure 1, contains the numbers we see in **Gifts received on day n** column of Table 1. These are the triangular numbers (Equation (1)). Similarly, the 4th diagonal of Pascal's triangle, shown in red in Figure 1, contains the **Total number of gifts received by day n** column of Table 1. These are the tetrahedral numbers (Equation (2)).

1
1 1
1 2 1
1 3 3 1
1 4 6 4 1
1 5 10 10 5 1
1 6 15 20 15 6 1
1 7 21 35 35 21 7 1
1 8 28 56 70 56 28 8 1
1 9 36 84 126 126 84 36 9 1
1 10 45 120 210 252 210 120 45 10 1
1 11 55 165 330 462 462 330 165 55 11 1
1 12 66 220 495 792 924 792 495 220 66 12 1
1 13 78 286 715 1287 1716 1716 1287 715 286 78 13 1
1 14 91 364 1001 2002 3003 3432 3003 2002 1001 364 91 14 1

Figure 1: Pascal's triangle with the triangular numbers in **blue** and the tetrahedral numbers in **red**

Pascal's triangle can also be drawn with binomial coefficients. This is shown is Figure 2.

$$\binom{0}{0}$$

$$\binom{1}{0} \quad \binom{1}{1}$$

$$\binom{2}{0} \quad \binom{2}{1} \quad \binom{2}{2}$$

$$\binom{3}{0} \quad \binom{3}{1} \quad \binom{3}{2} \quad \binom{3}{3}$$

$$\binom{4}{0} \quad \binom{4}{1} \quad \binom{4}{2} \quad \binom{4}{3} \quad \binom{4}{4}$$

$$\binom{5}{0} \quad \binom{5}{1} \quad \binom{5}{2} \quad \binom{5}{3} \quad \binom{5}{4} \quad \binom{5}{5}$$

$$\binom{6}{0} \quad \binom{6}{1} \quad \binom{6}{2} \quad \binom{6}{3} \quad \binom{6}{4} \quad \binom{6}{5} \quad \binom{6}{6}$$

$$\binom{7}{0} \quad \binom{7}{1} \quad \binom{7}{2} \quad \binom{7}{3} \quad \binom{7}{4} \quad \binom{7}{5} \quad \binom{7}{6} \quad \binom{7}{7}$$

$$\binom{8}{0} \quad \binom{8}{1} \quad \binom{8}{2} \quad \binom{8}{3} \quad \binom{8}{4} \quad \binom{8}{5} \quad \binom{8}{6} \quad \binom{8}{7} \quad \binom{8}{8}$$

$$\binom{9}{0} \quad \binom{9}{1} \quad \binom{9}{2} \quad \binom{9}{3} \quad \binom{9}{4} \quad \binom{9}{5} \quad \binom{9}{6} \quad \binom{9}{7} \quad \binom{9}{8} \quad \binom{9}{9}$$

$$\binom{10}{0} \quad \binom{10}{1} \quad \binom{10}{2} \quad \binom{10}{3} \quad \binom{10}{4} \quad \binom{10}{5} \quad \binom{10}{6} \quad \binom{10}{7} \quad \binom{10}{8} \quad \binom{10}{9} \quad \binom{10}{10}$$

$$\binom{11}{0} \quad \binom{11}{1} \quad \binom{11}{2} \quad \binom{11}{3} \quad \binom{11}{4} \quad \binom{11}{5} \quad \binom{11}{6} \quad \binom{11}{7} \quad \binom{11}{8} \quad \binom{11}{9} \quad \binom{11}{10} \quad \binom{11}{11}$$

$$\binom{12}{0} \quad \binom{12}{1} \quad \binom{12}{2} \quad \binom{12}{3} \quad \binom{12}{4} \quad \binom{12}{5} \quad \binom{12}{6} \quad \binom{12}{7} \quad \binom{12}{8} \quad \binom{12}{9} \quad \binom{12}{10} \quad \binom{12}{11} \quad \binom{12}{12}$$

$$\binom{13}{0} \quad \binom{13}{1} \quad \binom{13}{2} \quad \binom{13}{3} \quad \binom{13}{4} \quad \binom{13}{5} \quad \binom{13}{6} \quad \binom{13}{7} \quad \binom{13}{8} \quad \binom{13}{9} \quad \binom{13}{10} \quad \binom{13}{11} \quad \binom{13}{12} \quad \binom{13}{13}$$

$$\binom{14}{0} \quad \binom{14}{1} \quad \binom{14}{2} \quad \binom{14}{3} \quad \binom{14}{4} \quad \binom{14}{5} \quad \binom{14}{6} \quad \binom{14}{7} \quad \binom{14}{8} \quad \binom{14}{9} \quad \binom{14}{10} \quad \binom{14}{11} \quad \binom{14}{12} \quad \binom{14}{13} \quad \binom{14}{14}$$

Figure 2: Pascal's triangle with binomial coefficients

4 Geometric interpretations

There are interesting geometric interpretations of both the triangular and tetrahedral numbers. So first, what exactly is a triangular number? The first six triangular numbers are depicted as triangles in Figure 3. We can see from the figure that a triangular number T_n is the sum of the integers from 1 to n [6]. This is also the number of gifts received on the nth day of Christmas according to the "12 Days of Christmas".

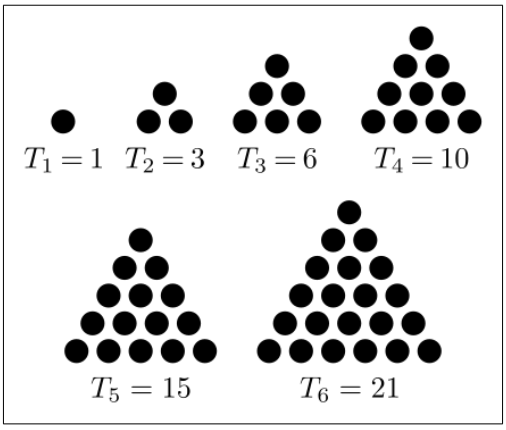

Figure 3: The first six triangular numbers

Ok, then what are the tetrahedral numbers? As we saw in Equation (2), the nth tetrahedral number is the sum of the first n triangular numbers [2]. If we stack up these triangular numbers we get a tetrahedron (hence the name "tetrahedral number"). For example, T_{e_4} is depicted in Figure 4. The nth tetrahedral number turns out to be the number of gifts received up to and including day **n** in "The 12 Days of Christmas".

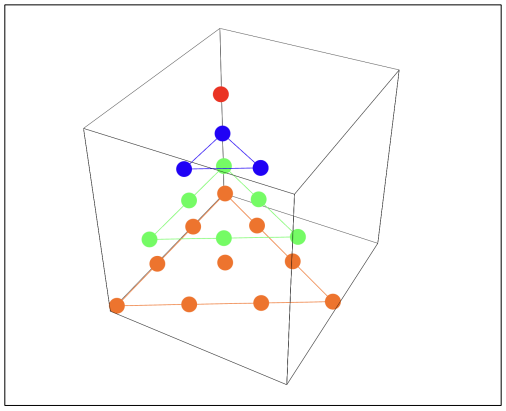

Figure 4: The fourth tetrahedral number $T_{e_4} = 20$

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